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Medication adherence among allogeneic hematopoietic stem cell transplanted patients: protocol of a systematic review

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Medication adherence among allogeneic hematopoietic stem cell transplanted patients: protocol of a systematic review

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ABSTRACT

Introduction Hematopoietic stem cell transplanted (HSCT) patients are subjected to complex oral medication based on prophylactic and immunosuppressive treatment. Adherence to medication plays a role in survival, and non-adherence (NA) is closely associated with Graft-versus-Host Disease (GvHD) and other complications. The aim of this review is to summarize the available evidence regarding medication adherence (ME), such as prevalence rates, risk factors of NA, effectiveness of interventions to support it and outcomes related to medication NA in this setting.

Methods and analysis A systematic review will be designed according to the Joanna Briggs Institute (JBI) methodology. CINAHL, the Cochrane Library, EMBASE, MEDLINE, PsychINFO and Scopus will be searched; also, studies from grey literature will be included. Primary studies on adult allogeneic HSCT patients who must take oral medication in the post-transplant phase will be included. The primary outcome will be the prevalence of oral ME; the secondary outcomes will be the risk factors of NA, the interventions aimed at increasing ME and the outcomes of NA, such as infections, acute and chronic GvHD, hospital readmission rate, disease relapse and overall survival. Two researchers will independently screen the eligible studies, then extract and produce the findings. Disagreements will be resolved by a third researcher. The risk of bias of individual included studies will be assessed using appropriate JBI Critical Appraisal Checklists.

Ethics and dissemination This comprehensive and quality assessed review will summarize the evidence about ME, providing prevalence rates, predictors, interventions to improve it and clinical outcomes among the adult HSCT population. The findings will be of interest to health care professionals working in the bone marrow transplantation setting, to patients, their caregivers and to representatives of patients. The review will be disseminated through conference presentations and will be published in international peer-reviewed scientific journals.

PROSPERO registration number CRD42022315298

Keywords: GvHD, Hematopoietic stem cell transplantation, Immunosuppressants, Infections, Interventions, Medication adherence, Medication non-adherence, Oral therapy, Prophylaxis, Risk factors

ARTICLE SUMMARY

Strengths and limitations of this study

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- The review will aim to summarize the available studies regarding the phenomenon of oral medication adherence respect to incidence, predictors, interventions to improve it and outcomes in the HSCT population, given that there is no comprehensive and up-to-date review
- The study will provide evidence data on this phenomenon increasing the HCPs' awareness of the educational needs of HSCT patients during hospitalization and over time
- A possible publication bias might limit the study, although the selection process will be conducted on six databases and consulting the gray literature

INTRODUCTION

The number of hematopoietic stem cell transplants (HSCTs) is increasing worldwide, with 48,512 transplants in the year 2019, including 19,798 allogeneic transplants (41%) in Europe,[1] and 23,768 HSCTs in the United States, of which 9,498 (40%) were allogeneic.[2] Bone marrow transplanted patients must take a complex oral medication regimen based upon prophylactic and immunosuppressive treatment.[3-5] Thus, patients may encounter difficulties managing the high number of tablets, which may vary over time due to medication prescription changes.[6] A high adherence to this highly complex regimen is required to avoid infections and to minimize the risk of other complications, such as Graft-versus-Host Disease (GvHD) or disease relapse.[7] It is well known that adherence plays a key role during and after the HSCT, increasing survival, and non-adherence (NA) is closely associated with GvHD.[4,6,8,9]

Adherence is defined by the World Health Organization (WHO) as *the extent to which the patient follows medical instructions*[10]. In particular, it is composed of three phases: initiation (starting the prescribed medication intake), implementation (the agreement between a patient's actual dosing compared to the medical prescription) and persistence (the length of time between the first and last dose taken).[11] Where compliance implies patient passivity, where the patient merely carries out the instructions given by health professionals without active involvement, adherence presumes the patient's agreement with the medical recommendations.[12,13] Therefore, adherence aims to render the patient or his/her relatives as active partners in the process of care.[13]

The adherence rate in developed countries is about 50% among patients with chronic illness.[13] To the best of our knowledge, the only systematic review available with regard to medication adherence among bone marrow recipients was published by Morrison and colleagues in 2017,[7] which included five studies, two on adults (a prospective cohort study and a prospective correlational study[14,15]), one on both paediatrics and

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adults (a phase II randomized controlled trial[16]) and two on paediatrics (a retrospective cohort study and a case study of longitudinal prospective dataset[17,6]). The estimated adherence to oral therapy was found to be between 33% and 94.7%.[7] However, medication adherence has been reported to decline over time among adults: in Hoodin *et al* (1993),[15] the overall rate of adherence, by self-assessment, was almost 95%, but the agreement of adherence rates between self-reporting measures and pill counting was 50%. In addition, the study by Martin *et al* (2012)[16] reported that adherence rates decreased from 53% to 40% and from 57% to 33% using pill counting. The only exception was the study by Chieng *et al* (2013),[14] where the mean adherence score using a modified Morisky questionnaire (total score from 0 – high adherence to 4 – low adherence) decreased by 1.53 (p = 0.0001) over six visits, showing an improvement in medication adherence.

In addition to the review published by Morrison *et al* (2017),[7] a prospective survey among 200 outpatient allogeneic HSCT adults was conducted in 2020 by Ice and colleagues[4] to determine the prevalence of NA to immunosuppressant and other medications. A total of 51% patients showed NA to nonimmunosuppressant medications, while 37.9% reported NA in taking oral immunosuppressants. The following cross-sectional multicentric survey enrolling 203 French adult allogeneic transplanted patients reported a NA rate of 75%, with an age of less than 50 years being a significant risk factor (p = 0.05).[18] Moreover, even a secondary data analysis investigating the patient-reported outcomes (PROs) of long-term survivors after allogeneic HSCT was conducted to assess medication NA in 99 patients, showing a prevalence rate of NA to immunosuppressants of 68.7%, 62.2% in patients with no/mild chronic GvHD and 80.2% in patients with moderate/severe chronic GvHD.[8]

Various methods have been used to date to assess adherence, which can be categorized into subjective, objective or biochemical measures.[12] Among the subjective approaches, self-reported questionnaires (ie, the Compliance Evaluation Test,[18] the modified Self-Efficacy for Appropriate Medication Use Scale,[19] the Immunosuppressant Therapy Adherence Scale[4] and the Morisky Medication Adherence Scale [4,7]) have been used. Among the objective measures, the electronic Medication Administration Record[19] and prescription refill records[4] were reported, while other studies used serum drug level measurements, of immunosuppressants in particular, alone[20] or in combination[4] to detect medication adherence. The use of self-reporting approaches in combination with objective methods has been recommended as the state-of-the-art adherence measurement.[7,12,13]

Several factors have been associated with NA among individuals with solid transplantation;[3] however, the phenomenon is still under investigated in bone marrow transplanted patients. Two studies have recently been conducted to seek psychosocial factors of NA in this population. Mishkin *et al* (2019)[21] assessed the

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Stanford Integrated Psychosocial Assessment for Transplant recipients (SIPAT), previously validated in the solid organ transplanted population,[22] to predict NA after HSCT: the NA rate was 21% (18 out of 85 patients), and the SIPAT score, treated as a continuous variable, was associated with NA (per 1 point; odds ratio (OR) 1.106; $p = 0.001$). In particular, a high psychosocial risk detected by the SIPAT was associated with increased NA, with a SIPAT score of 18 or above as a cut-off value to define "high-risk" of NA (specificity of 89.6% and sensitivity of 55.6%). Moreover, an increased risk of NA emerged among allogeneic HSCT with respect to autologous transplants (OR 14.184; $p = 0.005$).[21]

The second study, led by Scherer *et al* (2021),[9] used the Transplant Evaluation Rating Scale (TERS) as a pretransplant psychosocial screening instrument to predict mortality, and secondary, medication adherence and GvHD occurrence. A total of 61 allogeneic HSCT patients were prospectively followed in a 3-year follow-up, testing the predictive value of the TERS compared to mortality (higher TERS scores correlated with higher psychosocial impairment, on ten separate psychosocial domains, with risk stratification in low and increased); even medication adherence was associated with the TERS, measured using the Medication Experience Scale for Immunosuppressants (MESI). As emerged from the study findings, higher TERS scores were associated with lower survival for 90 days after HSCT (Hazard ratio (HR) 2.399); however, no correlation was seen between the MESI and TERS scores at the different data collection times, both in the inpatient (at three months after HSCT: $p = 0.85$, $r = -0.026$; at one year after HSCT: $p = 0.12$, $r = -0.206$) and outpatient setting (at three months after HSCT: $p = 0.96$, $r = -0.009$; at one year after-HSCT: $p = 0.9$, $r = 0.03$), respectively.

Even a higher number of daily immunosuppressant pills was associated with medication NA (OR 1.42; $p = 0.011$), as pointed out by Gresch *et al* (2017)[8] in their letter to the editor, where they reported the findings of a multivariate logistic regression of a secondary data analysis in 99 allogeneic HSCT recipients. In addition, the multivariate analysis led by Ice *et al* (2020)[4] in their cross-sectional survey reported that older age was associated with decreased odds of NA (per year, OR 0.97; $p = 0.016$), while the presence of distress, using the Distress Thermometer Score, was the only risk factor significantly associated with NA (OR 1.17; $p = 0.026$).[4] As emerged from the systematic review[7] and recommended by the Patient-Centered Outcomes Working Group of the National Institutes of Health in 2015,[23] future research is warranted to understand the facilitators and barriers of medication adherence.

Even interventions to improve medication adherence have only been investigated to a limited extend, and to our best knowledge no summary of the available studies have been published to date. Specifically, a French retro-prospective study[20] has recently been conducted to assess the effectiveness of pharmaceutical

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consultations – consisting of patient education, personalized medication intake schedules and proactive medication reconciliation – on the adherence to oral immunosuppressive therapy. The intervention was performed by a hospital pharmacist to allogeneic patients (n = 26) post-HSCT, the day before discharge from the HSCT unit and during weeks two and four after discharge; the historical cohort (n = 35) did not receive any pharmaceutical consultations. The study did not show significant findings in favour of the intervention group on medication adherence (rates of serum level assays in the therapeutic target range: 61.5% *vs.* 53.0%, p = 0.07), not even in relation to acute GvHD (53.8% *vs.* 51.4%, p = 0.85) and infections (26.9% *vs.* 22.8%, p = 0.72). In the quasi-experimental study by Polito *et al* (2021),[19] medication counselling by a pharmacist and self-management with nursing supervision until discharge was compared to a control group (medication counselling only at discharge without nursing supervision) among 51 allogeneic transplanted patients. No differences emerged among the two groups with regard to medication adherence (p = 0.12), although higher median knowledge scores were reported in the experimental group at discharge (8.5/10 *vs.* 10/10; p = 0.0023), using two *ad hoc* developed questionnaires.

The effects of medication adherence on clinical outcomes, such as infections, GvHD, disease relapse and mortality, are emerging in the literature.[4,8,9,20,21] Mishkin *et al* (2019)[21] used the SIPAT score to predict NA in 85 HSCT recipients and reported no significant associations between SIPAT ratings and survival; however, patients considered as having a high-risk SIPAT score had an increased risk of being admitted to the Intensive Care Unit (ICU) (OR 1.09; p = 0.021). Similarly, no differences emerged in survival between adherent and non-adherent patients in the prospective survey by Ice and colleagues (HR 1.43; p = 0.19),[4] nor with respect to acute GvHD (53.8% *vs.* 50.3%, p = 0.85) and infections (26.9% *vs.* 22.8%; p = 0.72) in the quasi-experimental trial by Charra *et al* (2021).[20] However, a difference in the development of mild chronic GvHD has been reported among non-adherent patients compared to adherent patients (OR 2.63; p = 0.042),[4] which was confirmed by the secondary analysis performed in 99 post-allogeneic HSCT patients, with a positive association between NA and higher grades of chronic GvHD (OR 3.01; p = 0.012).[8] Finally, a pre-transplant screening with the TERS contributed to the prediction of survival after HSCT; however, the reason why remains unclear, because the TERS did not correlate with the MESI, used as an adherence measure in 61 patients in the prospective study by Scherer *et al* (2021).[9]

Consequently, it can be seen that studies regarding oral medication adherence, its predictive factors, the effectiveness of interventions promoting it and the clinical associated outcomes among the adult allogeneic HSCT population are sparse and date back to recent years.[4,7-9,18-21] With the continuous progress of medication in this field, which might have increased the adherence, there is a need to perform an up-to-date systematic

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review. Accordingly, this review aims to provide a comprehensive and quality assessed summary of studies in order to inform the clinical practice and to address further primary research to cover the gaps that emerge.

Thus, the aim of the systematic review is to summarize the available studies regarding the adult allogeneic HSCT population with respect to:

a) the phenomenon of oral medication adherence, as prevalence rates and data collection tools used to measure it;

b) the predictors and risk factors of medication NA;

c) the effectiveness of interventions to promote medication adherence; and

d) the clinical outcomes influenced by medication NA.

METHODS AND ANALYSIS

The present protocol has been registered on the International Prospective Register of Systematic Reviews (PROSPERO) (registration number CRD42022315298, on 8th April 2022), and follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P) (see Supplementary table 1).[24] Therefore, this protocol provides the rationale for the systematic review, as well as the planned methodology to search and synthetise the findings. The review process will be completed within six months of protocol publication, and the systematic review will be reported following the updated Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement.[25]

Eligibility criteria

We will consider all primary studies to be eligible for the review, ensuring the following elements as defined with the PICO(T) framework, as recommended by the Joanna Briggs Institute (JBI) methodology:[26]

- *Population:* adult (age ≥ 18 years old) patients who have undergone allogeneic or haploidentical HSCT and must take oral medication in the post-transplant phase, after discharge from bone marrow transplantation centres (BMTCs), regardless of the number of medication pills to be taken.
- *Intervention*: all kind of interventions aimed at increasing medication adherence delivered from any health care professionals (HCPs), during hospitalization in the BMTCs and/or in the post-transplant phase, as an outpatient or remote follow-up.
- *Comparison*: usual care (eg, consultations, counselling).

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- *Outcome(s):* primary outcome describing the prevalence of oral medication adherence to immunosuppressant and nonimmunosuppressant medications among the selected population. Secondary outcomes are: (a) risk factors, barriers and predictors, and possibly related screening tools of NA; (b) effectiveness of interventions aimed at increasing medication adherence; and (c) infections (bacterial, viral and fungal), acute and chronic GvHD, hospital readmission rate, disease relapse and overall survival, as associated to medication NA.
- *Time:* from transplantation to three years post-HSCT (the longest follow-up emerged from the available studies[9]).

Inclusion and exclusion criteria

Our target population consists of adult patients who underwent allogeneic or haploidentical HSCT, for any transplantation indications, and must take oral medications (both immunosuppressants and nonimmunosuppressants, such as prophylactic medications) in the post-transplant phase.

We will include primary studies: (a) on humans of any publication year, (b) written in any language, (c) experimental (both randomized and non-randomized), observational (prospective and retrospective cohort and case-control), correlational, cross-sectional and longitudinal, in addition to case series and case reports, and d) with a low risk of bias, according to the quality assessment.[26] Therefore, secondary, qualitative studies and publications without original data (eg, comments, letters to the editor and editorials) or including the paediatric population (< 18 years old) or relating to autologous HSCT will all be excluded. Also, studies where adherence does not relate to medication (eg, diet regimen) will be excluded.

Primary and secondary outcomes

The primary outcome of the review is the oral medication adherence rate. We will adopt the theoretical framework proposed by the WHO (2003),[13] where adherence is defined as *the extent to which a person's behaviour – taking medication, following a diet, and/or executing lifestyle changes, corresponds with agreed recommendations from a health care provider*. However, the focus of adherence will be the medication regimen, thus excluding *a diet, and/or executing lifestyle changes* because they are not in line with our research aims. In order to measure medication adherence rate, subjective (ie, self-reporting questionnaires), objective (eg, prescription refill records) or biochemical measurements (serum drug levels) will be investigated.[12]

Secondary outcomes will include:

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(a) risk factors, barriers or predictors, as also measured with screening tools, of medication NA;

(b) interventions increasing medication adherence (eg, description, frequency of provision and duration);

(c) clinical outcomes, as a consequence of medication NA, such as the incidence rates of bacterial, viral and fungal infections, incidence rates of acute and chronic GvHD,[27] hospital and ICU readmission prevalence rates, disease relapse rates and overall survival rates.

Other outcomes that the review will aim to assess are the influences, if any (correlations/association), between medication adherence/NA and PROs, such as health-related quality of life, symptom experiences and patient satisfaction, as the most measured PROs in cancer literature.[28]

Information source

The following databases will be accessed: Cochrane Library, Cumulative Index of Nursing and Allied Health (CINAHL), EMBASE, MEDLINE via PubMed, PsychINFO and Scopus. An initial search in the PubMed database will be performed by the first researcher (XX) to identify keywords and Medical Subject Headings (MESH) appropriate to the aims of the review. Then, the search strategy will be refined and tested by the other researchers (XX, XX); thereafter, the search string will be inserted into the other databases.

We will also search for grey literature, including reports, protocols or guidelines from international scientific societies regarding medication adherence in the HSCT population. The reference lists of the included articles and the excluded systematic reviews will be screened using a hand search, in order to research additional sources. In the case of studies not available in their full text, the authors will be directly contacted.

Search strategy

Different search strings will be designed and applied, considering the primary outcome and the three secondary outcomes. The date of the last update of the literature search is 30th April 2022. An example of the initial search strategy in PubMed, with MeSH terms and keywords, is shown in Supplementary table 2.

Data management and selection process

Citations retrieved from the six electronic databases will be downloaded and imported into software (EndNote®) that will facilitate the study selection process, with automatic identification and exclusion of the duplicates.

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After the removal of duplicates, the first researcher (XX) will screen article titles and abstracts according to the inclusion and exclusion criteria; then, another researcher (XX) will independently screen the articles, reaching a consensus on eligible studies to include in the review. Disagreements between the researchers will be solved using consensus and consultation with a third researcher (XX), who will not have participated in the screening phase. A full-text assessment will then be performed by two researchers (XX, XX) independently, and then with common agreement with the third researcher (XX). The lists of references from the studies will also be screened, with the aim of retrieving and adding relevant studies, following the inclusion criteria. The selection process will be presented in the PRISMA flow-chart.[25]

Data collection process

A pilot-grid will be created in Microsoft Word® to extract data for at least 10% of the included articles, which will be conducted independently by the first (XX) and second researcher (XX). After the pilot test, changes will be included if necessary, and the final chart will be developed, where study data will be reported and double-checked. Discrepancies in data extraction by the researchers will be discussed until a consensus is reached, with the supervision of a third researcher (XX).

An example of information that will be extracted from each study could be:

- main characteristics (author(s), publication year, country, study design, study setting, study duration);
- main and secondary purpose(s);
- population (number, median age, gender distribution, haematology disease due to HSCT, type of allogeneic HSCT, time from HSCT, name of oral medication(s), divided into immunosuppressants or nonimmunosuppressants);
- primary outcome (prevalence rate of oral medication adherence, metrics and timing of adherence measurement);
- investigated risk factors, predictors, barriers (description), by also describing the screening tools for risk of medication NA, if in line with the study aims;
- investigated intervention(s) to improve medication adherence (description of the intervention, timing and provider(s) of the intervention(s)), if in line with the study aims;
- investigated secondary outcomes (eg, incidence rate of bacterial, viral and/or fungal infections, acute and/or chronic GvHD, hospital and/or ICU readmission rate, disease relapse rate and/or overall survival rate) and time from HSCT (in days) to assess them;

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- main results of the study.

Risk of bias

We will perform a risk of bias assessment of all the individual studies included. The assessment will be independently performed by two researchers (XX, XX), and conflicts will be solved by a third researcher (XX).

The assessment will be based on the use of the JBI Appraisal Checklists,[26] according to the study design. The JBI has separate checklists for every study design (eg, RCTs, quasi-experimental, cohort, cross-sectional and case-control studies, case series and case reports). An example is the Critical Appraisal Instrument for Studies Reporting Prevalence Data that will be used to assess the risk of bias of cross-sectional studies, which assesses: appropriateness of inclusion criteria (target population, sampling, sample size), description of participants and setting, use of valid and reliable measurement of the studied condition, use of appropriate statistical analysis, and assessment and management of adequate response rate.[29]

For every item, there could be four responses: 'yes', 'no', 'unclear' or 'not applicable'. The overall appraisal for each included study will consist of 'seek further info', 'include' or 'exclude' the study from the final inclusion.[26] Therefore, studies valued as excludable will not be considered in the data synthesis and a brief comment explaining the reason for the exclusion will indicated. A table will be provided with the results of the risk of bias assessment for each included study.

Data synthesis

Two researchers (XX, XX) will independently summarise the findings, including only those studies considered eligible after the risk of bias assessment. A qualitative narrative synthesis of the results will be provided; specifically, the findings emerged will be brought together in four different dimensions, according to the primary and the three secondary aims of the review. Within each aim, the main features of emerging outcomes (eg, prevalence rates, data collection methods/description of risk factors of NA, description of interventions) will be provided, in addition to other characteristics, such as the oral medication, the study design and its duration, the sample size, the participants (eg, age, gender, type of HSCT, the author(s) and the publication year of the study).

Regarding the investigated outcomes, due to the different medication adherence measures and interventions across the studies, we acknowledge that the heterogeneity of the identified studies may impede the ability to conduct a meta-analysis. However, if the data allows, two researchers (XX, XX) will independently

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investigate the heterogeneity of the studies using the I^2 statistic. If a meta-analysis will not be possible, we will provide a descriptive synthesis of the findings, as above.

Confidence in cumulative evidence

The quality of evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE] framework[30] for both primary and secondary outcomes. 'High', 'moderate', 'low' and 'very low' are the levels that will be used to assess the strength and quality of the evidence regarding the identified outcomes. The evaluation will be independently performed by two researchers (XX, XX), and an expert researcher (XX) will resolve any disagreements.

ETHICS AND DISSEMINATION

Since the study will be merely based on published and retrievable literature, no ethics approval will be obtained. This review aims to summarize the available studies regarding the phenomenon of oral medication adherence by means of prevalence rates, predictors and risk factors of NA, effectiveness of interventions to support it and outcomes associated with medication NA in the HSCT population, given that the dates of most of the available studies are not recent and there is no comprehensive and up-to-date review. The findings will be of interest to various stakeholders, including HCPs (such as nurses, haematologists and psychologists working in the BMTCs and in the follow-up setting), bone marrow transplanted patients, their caregivers and their representatives, as associations. Although disease relapse remains the leading cause of mortality among post-transplanted bone marrow patients, a substantial improvement in survival and a reduction of complications have been reported over the years.[31] Therefore, also considering the increase of GvHD incidence,[8] medication adherence has become a priority in designing and delivering health care interventions during and after HSCT.[4,6,8,9]

Complications or failure of transplant during hospitalization are mostly attributed to biological reasons.[9] Medication adherence among transplanted patients increases its relevance later, especially in the case of early discharge and after, in the outpatient follow-up. Providing evidence data on the phenomenon of medication adherence might increase the HCPs' awareness of the educational needs during hospitalization and over time, since not all BMTCs have reference HPCs in charge of transitional care. Moreover, providing evidence might also increase the opportunities to benchmark the data at each centre with that accumulated in the literature and appropriately summarised.

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The findings of our systematic review will be submitted to international peer-reviewed scientific journals and organised in the identified primary and secondary outcomes. Furthermore, the summary of the systematic review will be disseminated through professional meetings and conference presentations, to promote implementation in the HCPs' practice, such as providing screening tools to recognize barriers and risk factors of medication NA, or interventions to improve medication adherence. Finally, the findings of this review may also reveal possible gaps in the HSCT literature and highlight primary research questions.

Patient and public involvement

Patients or the public were not involved in the design of the protocol.

Author contributions

CV and AP conceptualized the protocol. CV wrote the protocol. IM and AP methodologically and critically revised the protocol. CV is the guarantor of the review. All authors have contributed, read and approved the final protocol.

Competing interests

None declared.

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Data Availability Statement'

Not applicable.

Patient consent for publication

Not required.

Ethical approval

Not required.

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Supplementary table 1 Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P) checklist: recommended items to address in a systematic review protocol[24]

Section/topic	#	Checklist item	Information reported		Page n.
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	Yes		1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		No	-
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number in the Abstract	Yes		2
Authors					
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Yes		1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Yes		13
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		No	-
Support					
Sources	5a	Indicate sources of financial or other support for the review		No	-
Sponsor	5b	Provide name for the review funder and/or sponsor		No	-
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		No	-
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	Yes		3
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Yes		7
METHODS					
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Yes		7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Yes		9
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Yes		9
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Yes		9

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Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Yes		10
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Yes		10
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Yes	No	-
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Yes		8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Yes		11
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	Yes		11
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Yes		11
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)		No	-
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Yes		12
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)		No	-
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Yes		12

n, number.

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Supplementary table 2 Draft of the search strategy for the MEDLINE database

Investigated Outcome(s)	Search Terms
Medication adherence/NA	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”)
Medication NA and risk factors	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Risk Factors"[Mesh] OR risk factor OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening tool OR psychosocial assessment* OR evaluation* OR psychosocial*)
Interventions increasing medication adherence	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Psychosocial Intervention"[Mesh] OR intervention* OR psychological intervention* OR education*)
Medication adherence/NA and GvHD	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Graft vs Host Disease"[Mesh])
Medication adherence/NA and infections	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Infections"[Mesh] OR viral infection* OR bacterial infection* OR fungal infection*)

Medication adherence/NA and hospital readmission	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Patient Readmission"[Mesh] OR hospital readmission* OR intensive care unit admission*)
Medication adherence/NA and survival	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Survival"[Mesh] OR "Mortality"[Mesh])
Medication adherence/NA and disease relapse	("Hematopoietic Stem Cell Transplantation"[Mesh] OR Stem Cell Transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR allogeneic) AND ("Medication Adherence"[Mesh] OR Adherence Medication OR Drug Adherence OR Medication Nonadherence OR Medication Non-Adherence OR Medication Persistence) AND (Pharmaceutic* OR Pharmaceutical Preparation* OR Pharmaceutical Product* OR Drug* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR “oral therapy”) AND ("Recurrence"[Mesh] OR recurrence* OR recrudescence* OR relapse* OR disease relapse*)

GvHD, Graft-versus-Host Disease; NA, non-adherence.

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Medication adherence among allogeneic haematopoietic stem cell transplant recipients: a systematic review protocol

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ABSTRACT

Introduction Patients receiving a haematopoietic stem cell transplant (HSCT) are subjected to complex oral medications based on prophylactic and immunosuppressive treatments. Adherence to medication plays a role in survival, and medication non-adherence (MNA) is closely associated with graft-versus-host disease (GvHD) and other complications. The aim of this systematic review is to summarise the available evidence regarding prevalence rates of medication adherence, the risk factors of MNA, the effectiveness of interventions to increase medication adherence and the outcomes associated with MNA.

Methods and analysis We designed a systematic review according to the Joanna Briggs Institute methodology. We will search the Cochrane Library and the CINAHL, EMBASE, MEDLINE via PubMed, PsychINFO and Scopus databases. We will include published and unpublished primary studies: (a) on humans, from inception until 10 May 2022; (b) written in any language; (c) experimental (randomised and non-randomised), observational (prospective, retrospective cohort and case-control), correlational, cross-sectional and longitudinal; and (d) with a low risk of bias, according to the quality assessment we perform. We will exclude secondary and qualitative studies, protocols, publications without original data, including paediatrics or related to autologous HSCT. The primary outcome will be the prevalence of oral medication adherence; the secondary outcomes will be the risk factors of MNA, the interventions aimed at increasing medication adherence and the outcomes of MNA. Two researchers will independently screen the eligible studies, then extract and describe the data. Disagreements will be resolved by a third researcher. We will provide a qualitative narrative synthesis of the findings.

Ethics and dissemination Ethical approval is not required because we will use previously published studies. The findings will increase the awareness of health care providers (HCPs), patients, their caregivers, and representatives in understanding the factors hindering or promoting medication adherence. Moreover, this review will inform HCPs regarding the effective interventions capable of increasing medication adherence. We will disseminate the findings through conference presentations and publications in international peer-reviewed scientific journals.

PROSPERO registration number CRD42022315298

Keywords: Graft-versus-host disease, Haematopoietic stem cell transplantation, Immunosuppressants, Infections, Interventions, Medication adherence, Medication non-adherence, Oral therapy, Prophylaxis, Risk factors

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ARTICLE SUMMARY

Strengths and limitations of this study

- The review will examine six databases and grey literature, leading to a broad synthesis of the available evidence regarding medication adherence among adult recipients of an allogeneic haematopoietic stem cell transplant.
- The review will include primary studies without language and time restrictions, allowing the maximum inclusion of studies published in the field.
- Studies with low methodological quality will be excluded, thus ensuring rigor and robustness in the findings of the review.
- The independence of the researchers in the selection, analysis and summary of the studies will ensure rigor and strengthen the findings of the review.
- According to the expected heterogeneity in the available studies, a quantitative synthesis of the results in the form of meta-analysis may not be possible.

INTRODUCTION

The number of haematopoietic stem cell transplants (HSCTs) is increasing worldwide, with 48,512 transplants in the year 2019, including 19,798 allogeneic transplants (41%) in Europe,[1] and 23,768 HSCTs in the United States, of which 9,498 (40%) were allogeneic.[2] Patients who receive a bone marrow transplant must take a complex oral medication regimen based upon prophylactic and immunosuppressive treatment.[3-5] Thus, patients may encounter difficulties managing the high number of tablets, which may vary over time due to medication prescription changes.[6] However, a high level of adherence to this very complex regimen is required to avoid infections and to minimise the risk of other complications, such as graft-versus-host disease (GvHD) or disease relapse.[7] It is well known that adherence plays a key role during and after the HSCT, increasing survival, and medication non-adherence (MNA) is closely associated with GvHD.[4,6,8,9]

Adherence is defined by the World Health Organization (WHO) as 'the extent to which the patient follows medical instructions'.[10] It is composed of three phases: initiation (starting the prescribed medication intake), implementation (the agreement between a patient's actual dosing compared with the medical prescription) and persistence (the length of time between the first and last dose taken).[11] While compliance

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implies patient passivity, adherence presumes the patient agrees with the medical recommendations[12,13] and is actively involved in the care process.[13]

The adherence rate in developed countries is about 50% among patients with chronic illness.[13] To the best of our knowledge, the only systematic review available with regard to medication adherence among bone marrow recipients was published by Morrison and colleagues in 2017.[7] The authors included five studies, two in adults,[14,15] one in both children and adults[16] and two in paediatric population. [6,17] The estimated adherence to oral therapy in that review was between 33% and 94.7%.[7] Adherence to oral therapy has been reported to decline over time among adults,[15,16] with the exception of Chieng *et al.*,[14] who reported an improvement in medication adherence over time. More recently, a prospective survey conducted by Ice and colleagues[4] showed that out of 200 patients, 51% of them were not adherent to non-immunosuppressant medications, and 37.9% were not adherent to oral immunosuppressants. In a cross-sectional multicentric study enrolling 203 French adult recipients of an allogeneic HSCT, the MNA rate was 75%.[18] In addition, in a secondary analysis of long-term survivors after an allogeneic HSCT, the authors reported a MNA prevalence rate of 68.7% to immunosuppressants.[8]

Various methods have been used to assess medication adherence, which can be categorised into subjective, objective or biochemical measures.[12] Among the first category, self-reported questionnaires (i.e. the Compliance Evaluation Test,[18] the modified Self-Efficacy for Appropriate Medication Use Scale,[19] the Immunosuppressant Therapy Adherence Scale[4] and the Morisky Medication Adherence Scale[4,7]) have been used. Among the objective measures, the electronic Medication Administration Record[19] and the prescription refill records[4] have been adopted by researchers, while among biochemical measures, the serum drug level measurements of immunosuppressants, alone[20] or in combination, have been used.[4] According to the available evidence, the use of self-report approaches combined with objective methods has been recommended[7,12,13] to prevent heterogeneity in prevalence rates.[21] Moreover, to increase the methodological quality of the studies, it has been recommended to report when adherence is measured, the medications prescribed and the conceptual definition of adherence considered by the researchers.[21]

The WHO[13] has classified reasons for poor medication adherence in chronic illnesses into five categories: socioeconomic factors, health system/health care team–related factors, disease-related factors, therapy-related factors and patient-related factors. Among individuals who have undergone solid transplantation, several factors have been investigated regarding their association with MNA.[3] At the patient-related level, some studies have documented the role of psychosocial factors[9,22] by using the Stanford Integrated Psychosocial Assessment for Transplant recipients (SIPAT), previously validated in the population

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who had received a solid organ transplant,[22,23] and the Transplant Evaluation Rating Scale (TERS) as a pretransplant screening instrument to predict mortality, medication adherence and GvHD occurrence. Patient age,[4] education,[24] distress[4] as well as the efficacy of the caregiver(s)[25] have also been investigated. At the therapy-related level, the long-term duration of the therapy[25] as well as the higher number of daily immunosuppressant pills have been studied.[8] However, as has emerged from a published systematic review[7] and recommended by the Patient-Centered Outcomes Working Group of the National Institutes of Health in 2015,[26] future research is warranted to understand the facilitators and barriers of medication adherence.

Interventions to improve medication adherence have only been investigated to a limited extent, and to our best knowledge no summary of the available studies has been published. Specifically, a recent French retro-prospective study[20] has assessed the effectiveness of pharmaceutical consultations on the adherence to oral immunosuppressive therapy. The intervention was performed by a hospital pharmacist to patients who had received an allogeneic HSCT (n = 26) the day before discharge from the HSCT unit and from 2 to 4 weeks after discharge. There were no significant findings in favour of the intervention group on medication adherence. In their quasi-experimental study, Polito *et al.*[19] compared medication counselling by a pharmacist and self-management with nursing supervision until discharge to a control group among 51 recipients of an allogeneic HSCT. There were no differences between the two groups regarding medication adherence, although at discharge the experimental group had higher median knowledge scores. Moreover, evidence regarding different educational strategies, such as peer education or the involvement of patients as trainers, has not yet been summarised.

The effects of medication adherence on clinical outcomes, such as infections, GvHD, disease relapse and mortality, are emerging in the literature.[4,8,9,20,22] Mishkin *et al.*[22] reported no significant associations between SIPAT ratings and survival; however, patients with a high-risk SIPAT score had an increased risk of being admitted to the intensive care unit (ICU). Similarly, no differences emerged in survival between adherent and non-adherent patients in the prospective survey by Ice and colleagues,[4] nor with respect to acute GvHD and infections in the quasi-experimental trial by Charra *et al.*[20] On the other hand, there was a difference in the development of mild chronic GvHD among non-adherent patients compared with adherent patients.[4] While pre-transplant screening with the TERS has been documented to contribute to prediction of survival after HSCT, the TERS score did not correlate with the MESI – used as an adherence measure – suggesting that mortality is not associated with MNA.[9] There is a need to summarise the evidence regarding clinical outcomes to discriminate between adherence and non-adherence behaviour and to identify patients at increased risk[21] and in need of tailored interventions.

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Studies regarding oral medication adherence prevalence, its associated factors, the effectiveness of interventions promoting it and the associated clinical outcomes among the adult recipients of an allogeneic HSCT are sparse and date back to recent years.[4,7-9,18-20,22] Therefore, this review is intended to provide a comprehensive and quality-assessed summary of studies to inform clinical practice and to address further primary research. Our primary aim is to summarise the available evidence regarding the phenomenon of oral medication adherence among adult recipients of an allogeneic HSCT based on prevalence rates and data collection tools used to measure it. Our secondary aim is to summarise the evidence regarding the predictors and risk factors of MNA, the effectiveness of interventions to promote medication adherence, and the clinical outcomes influenced by MNA, as documented in primary studies to date.

METHODS AND ANALYSIS

We have registered the present protocol in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number CRD42022315298, on 8 April 2022). This protocol follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P) (see Supplementary Table 1).[27] The review process will be completed within 6 months of protocol registration, and the systematic review will be reported following the updated Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement.[28]

Eligibility criteria

We will consider all primary studies to be eligible for the review, ensuring the following elements as defined with the Population, Intervention, Comparison and Outcomes and Time – PICO(T) – framework, as recommended by the Joanna Briggs Institute (JBI) methodology[29]:

- *Population* – adult (age ≥ 18 years old) patients who have undergone allogeneic or haploidentical HSCT and must take oral medication in the post-transplant phase, after discharge from bone marrow transplantation centres (BMTCs), regardless of the number of pills they must take;
- *Intervention* – all kinds of interventions aimed at increasing medication adherence delivered from any health care professionals (HCPs), during hospitalisation in the BMTCs and/or in the post-transplant phase, as an outpatient or remote follow-up;

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- *Comparison* – usual care (i.e. consultations, counselling, peer-education and/or no comparison group);
- *Outcome(s)* – primary outcome describing the prevalence of oral medication adherence to immunosuppressant and non-immunosuppressant medications among the selected population after discharge from BMTCs. The secondary outcomes are (a) risk factors, barriers and predictors, and possibly related MNA assessment tools; (b) effectiveness of interventions aimed at increasing medication adherence; and (c) infections (bacterial, viral and fungal), acute and chronic GvHD, hospital readmission rate, disease relapse and overall survival, as associated with MNA;
- *Time* – from transplantation to 4 years post-HSCT, according to the longest median follow-up among the available studies.[8]

Inclusion and exclusion criteria

Our target population consists of adult patients who have undergone allogeneic or haploidentical HSCT, for any transplantation indications, and must take oral medications (both immunosuppressants and non-immunosuppressants, such as prophylactic medications) in the post-transplant phase.

We will include all published and unpublished primary studies: (a) on humans, from inception until 10 May 2022; (b) written in any language; (c) experimental (randomised and non-randomised), observational (prospective, retrospective cohort and case-control), correlational, cross-sectional and longitudinal, in addition to case series and case reports; and d) with a low risk of bias, according to the quality assessment findings.[29] Therefore, we will exclude secondary studies, qualitative studies, study protocols, case series and case reports and publications without original data (i.e. comments, letters to the editor and editorials) or including the paediatric population (< 18 years old) or relating to autologous HSCT. In addition, we will exclude studies where adherence does not relate to medication (i.e. diet regimen, physical exercise).

Primary and secondary outcomes

The primary outcome of the review is the oral medication adherence rate. We will adopt the theoretical framework proposed by the WHO,[13] where adherence is defined as 'the extent to which a person's behaviour – taking medication, following a diet, and/or executing lifestyle changes, corresponds with agreed recommendations from a health care provider'. However, the focus of adherence will be the medication

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regimen, thus excluding 'a diet, and/or executing lifestyle changes' because they are not in line with our research aims. To measure the medication adherence rate, we will investigate subjective (i.e. self-report questionnaires), objective (e.g. prescription refill records) or biochemical measurements (serum drug levels).[12]

The secondary outcomes of this review are the following:

(a) risk factors, barriers or predictors of MNA as measured with tools;

(b) interventions increasing medication adherence (i.e. full description, frequency of provision and duration, comparator);

(c) clinical outcomes as associated with MNA, such as the incidence rates of bacterial, viral and fungal infections; the incidence rates of acute and chronic GvHD[30]; hospital and ICU readmission prevalence rates; disease relapse rates; and overall survival rates. We will assess correlations or associations, if any, between medication adherence/MNA and patient-reported outcomes (i.e. health-related quality of life, symptom experiences and patient satisfaction).[31]

Information source

We will access the following databases: the Cochrane Library, Cumulative Index of Nursing and Allied Health (CINAHL), EMBASE, MEDLINE via PubMed, PsychINFO and Scopus. An initial search in the PubMed database will be performed by the first researcher (CV) to identify keywords and Medical Subject Headings (MESH) appropriate to the aims of the review. Then, the search strategy will be refined and tested by the other researchers (IM, AP). Thereafter, the search string will be inserted into the other databases.

We will also search for grey literature, including dissertation theses and reports, protocols or guidelines from international scientific societies regarding medication adherence in the HSCT population. We will screen the reference lists of the included articles and the excluded systematic reviews using a hand search, to identify additional sources. If the full text of a study is not available, we will contact the authors directly.

Search strategy

Different search strings have been designed and applied, considering the primary outcome and the three secondary outcomes. The date of the last update of the literature search was 10 May 2022. The final search strategies for the six selected databases, with MeSH terms and keywords, are shown in Supplementary Table 2. No filters and/or limits have been used.

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Data management and selection process

We will download citations retrieved from the six electronic databases and import them into software (EndNote®) that will facilitate the study selection process, with automatic identification and exclusion of the duplicates.

After the removal of duplicates, the first researcher (CV) will screen article titles and abstracts according to the inclusion and exclusion criteria; then, another researcher (IM) will independently screen the articles, reaching a consensus on eligible studies to include in the review. Disagreements between the researchers will be solved using consensus and consultation with a third researcher (AP), who will not have participated in the screening phase. A full-text assessment will then be performed by two researchers (CV, IM) independently, and then with common agreement with the third researcher (AP). The lists of references from the studies will also be screened, with the aim of retrieving and adding relevant studies, following the inclusion criteria. We will present the selection process in the PRISMA flow-chart.[28]

Data collection process

A pilot-grid will be created in Microsoft Word® to extract data for at least 10% of the included articles, conducted independently by the first (CV) and second (IM) authors. After the pilot test, changes will be included if necessary, and the final chart will be developed, where study data will be reported and double-checked. Discrepancies in data extraction by the researchers will be discussed until a consensus is reached, with the supervision of a third researcher (AP).

An example of information that will be extracted from each study could be:

- main characteristics (author[s], publication year, country, study design, study setting, study duration);
- main and secondary study aims(s);
- population profile, such as the number of participants, median age, gender distribution, race, ethnicity, socioeconomic status (where available), haematological disease due to HSCT, type of allogeneic HSCT, time from HSCT and name of oral medication(s), categorised into immunosuppressants or non-immunosuppressants;
- main results of the study according to the primary and the secondary outcome of the review:
  (a) risk factors, predictors, barriers (description), by also describing the tools used to assess the risk of MNA;

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(b) intervention(s) to improve medication adherence (description of the intervention, timing and provider[s] of the intervention[s], comparator);

(c) outcomes (i.e. incidence rate of bacterial, viral and/or fungal infections, acute and/or chronic GvHD, hospital and/or ICU readmission rate, disease relapse rate and/or overall survival rate) and time from HSCT (in days) to assess them.

Risk of bias

We will perform a risk of bias assessment of all the individual studies included. The assessment will be performed independently by two researchers (CV, IM), and conflicts will be solved by a third researcher (AP).

The assessment will be based on the use of the JBI Appraisal Checklists,[29] according to the study design of the included studies. The JBI has different checklists for each study design (i.e. randomised controlled trials, quasi-experimental, cohort). An example is the Critical Appraisal Instrument for Studies Reporting Prevalence Data that will be used to assess the risk of bias of cross-sectional studies, which assesses the appropriateness of inclusion criteria (target population, sampling and sample size), the description of participants and the setting, the use of valid and reliable measurement of the studied condition, the use of appropriate statistical analysis, and the assessment and management of adequate response rate.[32]

There are four possible responses for each item: 'yes', 'no', 'unclear' or 'not applicable'. The overall appraisal for each included study will consist of 'seek further info' or 'include' or 'exclude' the study from the final inclusion.[29] Therefore, studies assessed as excludable will not be considered in the data synthesis and a brief comment explaining the reason for the exclusion will be indicated. We will provide a table reporting the results of the risk of bias assessment for each included study.

Data synthesis

Two researchers (CV, IM) will independently summarise the findings, including only those studies considered eligible after the risk of bias assessment. First, the main study features will be summarised by reporting, for example, the origin country of the authors, the study aims and design, its duration, the sample size, and the participants (i.e. age, gender, ethnicity, education, type of HSCT). Then, a qualitative narrative synthesis of the results will be provided; specifically, the findings that emerge will be synthesised in four dimensions, according to the primary and the three secondary aims of this review. For example, the prevalence rates reported in the studies for immunosuppressant and non-immunosuppressant medications will be presented

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with the minimum, maximum and median values. The data collection methods, the description of risk factors of MNA and the interventions applied to improve adherence will also be summarised.

Regarding the investigated prevalence and outcomes, due to the expected different medication adherence measures and interventions across studies,[7] the heterogeneity of the identified studies may prevent the ability to conduct a meta-analysis. However, if the data allow, two researchers (CV, IM) will independently investigate the heterogeneity of the studies using the I^2 statistic.

Confidence in cumulative evidence

We will assess the quality of the evidence by using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework[33] for both primary and secondary outcomes. 'High', 'moderate', 'low' and 'very low' are the levels that will be used to assess the strength and quality of the evidence regarding the identified outcomes. The evaluation will be performed independently by two researchers (CV, IM), and an expert researcher (AP) will resolve any disagreements.

Patient and public involvement

Patients and/or the public will not be involved in the design, conduct and reporting of this review.

ETHICS AND DISSEMINATION

Because the study will only be based on published and retrievable literature, no ethics approval is required. We will submit the findings of our systematic review to international peer-reviewed scientific journals organised based on the identified primary and secondary outcomes. Furthermore, we will disseminate a summary of the systematic review through professional meetings and conference presentations, to promote implementation in HCP practice, such as providing screening tools to recognise patients at risk, barriers and risk factors of MNA or interventions to improve medication adherence. Finally, the findings of this review may also reveal possible gaps in the HSCT literature and highlight primary research questions.

DISCUSSION

We aim to summarise the available studies regarding the phenomenon of oral medication adherence by means of prevalence rates, predictors and risk factors of MNA, the effectiveness of interventions to support it and outcomes associated with MNA among the HSCT population. Although we will employ a systematic approach

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and search an extensive number of databases, a publication bias might occur. Moreover, the expected heterogeneity among the studies regarding their measures, tools and interventions[7] might prevent a quantitative synthesis of the results.

The findings will be of interest to various stakeholders, including HCPs (such as nurses, haematologists and psychologists working in BMTCs and in the follow-up setting), patients who have received a bone marrow transplant, their caregivers and their representatives. Although disease relapse remains the leading cause of mortality among patients after receiving a bone marrow transplant, a substantial improvement in survival and a reduction of complications have been reported over the years.[34] Therefore, also considering the increase in the incidence of GvHD,[8] medication adherence has become a priority in designing and delivering health care interventions during and after HSCT.[4,6,8,9]

Complications or failure of transplant during hospitalisation have mostly been attributed to biological reasons.[9] There has been a recent increase in attention towards medication adherence among patients who have received a transplant, especially in the case of early discharge and after, in the outpatient follow-up. Therefore, the findings will increase the information available to HCPs, patients, their caregivers and representatives, in understanding the factors hindering or promoting medication adherence. Moreover, this review will inform HCPs regarding the most effective interventions capable of promoting medication adherence by considering all factors involved – from those related to the complexity of the therapy to the uncertainty and psychological distress experienced by the patients and their caregivers.[35] Providing data on the phenomenon of medication adherence might increase the awareness of HCPs regarding the educational needs during hospitalisation and over time, because not all BMTCs have reference HPCs in charge of transitional care. Moreover, providing evidence might increase the opportunities to benchmark the data at each centre with that accumulated in the literature and appropriately summarised.

Author contributions

CV and AP conceptualized the protocol. CV wrote the protocol. IM and AP methodologically and critically revised the protocol. CV is the guarantor of the review. All authors have contributed, read and approved the final protocol.

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Competing interests

None declared.

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Patient consent for publication

Not required.

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Supplementary table 1 Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P) checklist: recommended items to address in a systematic review protocol.[27]

Section/topic	#	Checklist item	Information reported		Section where item is reported
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	Yes		Title
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		No	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number in the Abstract	Yes		Abstract and Methods and Analysis
Authors					
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Yes		First page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Yes		Author contribution
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		No	
Support					
Sources	5a	Indicate sources of financial or other support for the review		No	
Sponsor	5b	Provide name for the review funder and/or sponsor		No	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		No	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	Yes		Introduction
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Yes		Eligibility criteria
METHODS					
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Yes		Inclusion and exclusion criteria
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Yes		Information sources
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Yes		Search strategy
STUDY RECORDS					

Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Yes		Data management and selection process
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Yes		Data management and selection process
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Yes		Data collection process
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications		No	
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Yes		Primary and secondary outcomes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Yes		Risk of bias
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	Yes		Data synthesis
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Yes		Data synthesis
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)		No	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Yes		Data synthesis
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)		No	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Yes		Confidence in cumulative evidence

n, number.

Supplementary table 2 Final search strategy for the six investigated databases (no limits and/or filters have been used).

Database	String number	Researched outcome	Search strings
PUBMED	#1	Medication adherence/MNA	("Hematopoietic Stem Cell Transplantation"[Mesh] OR "Transplantation, Homologous"[Mesh] OR stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND ("Medication Adherence"[Mesh] OR medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR "oral therapy" OR pill* OR tablet* OR capsule OR "Immunosuppressive Agents"[Mesh] OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND ("Risk Factors"[Mesh] OR risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR "screening tool" OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND ("Psychosocial Intervention"[Mesh] OR intervention* OR psychological intervention* OR "Patient Education as Topic"[Mesh] OR "patient education" OR education OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND ("Graft vs Host Disease"[Mesh] OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND ("Infections"[Mesh] OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND ("Patient Readmission"[Mesh] OR patient readmission* OR unplanned readmission OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND ("Survival"[Mesh] OR "Mortality"[Mesh] OR survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)
	#8	Medication adherence/MNA and disease relapse	#1 AND ("Recurrence"[Mesh] OR recurrence* OR recrudescence* OR relapse* OR disease relapse*)
CINAHL	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant* OR (MH "Bone Marrow Transplantation") OR (MH "Bone Marrow Transplantation, Allogeneic")) AND ((MH "Medication Compliance") OR (MH "Compliance with Medical Regimen (Saba CCC)") OR (MH "Compliance with Medication Regimen (Saba CCC)" OR medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR "oral therapy" OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR (MH "Immunosuppressive Agents") OR (MH "Immunosuppression"))

	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR "screening tool" OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier* OR (MH "Risk Factors") OR (MH "Social Factors"))
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication OR (MH "Psychosocial Intervention") OR (MH "Nursing Interventions") OR (MH "Intervention Trials") OR (MH "Patient Education") OR (MH "Counseling"))
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease* OR (MH "Graft Versus Host Disease"))
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis* OR (MH "Sepsis") OR (MH "Infection"))
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit* OR (MH "Readmission") OR (MH "Intensive Care Units"))
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival OR (MH "Survival") OR (MH "Mortality") OR (MH "Death"))
	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse* OR (MH "Recurrence"))
Cochrane Library	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND (medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR "oral therapy" OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR non immunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR "screening tool" OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)
	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse*)
PsychINFO	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND (medication

			adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR oral therapy OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR non immunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR screening tool OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)
	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse*)
SCOPUS	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND (medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR “oral therapy” OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR non immunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR “screening tool” OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)

	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse*)
EMBASE	#1	Medication adherence/MNA	('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw)
	#2	MNA and risk factor	('risk factor*':ti,ab,kw OR 'social risk factor*':ti,ab,kw OR 'health correlate*':ti,ab,kw OR 'risk score*':ti,ab,kw OR 'risk factor score*':ti,ab,kw OR predictor*:ti,ab,kw OR screening:ti,ab,kw OR 'screening tool':ti,ab,kw OR 'psychosocial assessment*':ti,ab,kw OR evaluation*:ti,ab,kw OR psychosocial*:ti,ab,kw OR facilitate*:ti,ab,kw OR barrier*:ti,ab,kw) AND ('pre transplant*':ti,ab,kw OR 'stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw)
	#3	Interventions increasing medication adherence	('psychosocial intervention*':ti,ab,kw OR intervention*:ti,ab,kw OR 'psychological intervention*':ti,ab,kw OR 'patient education':ti,ab,kw OR education*:ti,ab,kw OR consultation*:ti,ab,kw OR counselling*:ti,ab,kw OR implementation:ti,ab,kw OR 'self medication':ti,ab,kw) AND ('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw)
	#4	Medication adherence/MNA and GvHD	(('stem cell'/exp OR 'stem cell' OR (('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'bone marrow'/exp OR 'bone marrow' OR (('bone'/exp OR bone) AND ('marrow'/exp OR marrow) AND transplant*) OR 'haematopoietic stem cell'/exp OR 'haematopoietic stem cell' OR (haematopoietic AND

		('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'hematopoietic stem cell'/exp OR 'hematopoietic stem cell' OR (hematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR allogeneic) AND transplant* OR haploidentical) AND transplant* AND ('medication adherence'/exp OR 'medication adherence' OR (('medication'/exp OR medication) AND ('adherence'/exp OR adherence)) OR 'adherence medication' OR (('adherence'/exp OR adherence) AND ('medication'/exp OR medication)) OR 'medication nonadherence' OR (('medication'/exp OR medication) AND nonadherence) OR 'medication non-adherence' OR (('medication'/exp OR medication) AND 'non adherence') OR 'medication non adherence' OR (('medication'/exp OR medication) AND non AND ('adherence'/exp OR adherence)) OR 'drug adherence'/exp OR 'drug adherence' OR (('drug'/exp OR drug) AND ('adherence'/exp OR adherence)) OR 'medication persistence'/exp OR 'medication persistence' OR (('medication'/exp OR medication) AND ('persistence'/exp OR persistence)) OR 'medication noncompliance' OR (('medication'/exp OR medication) AND noncompliance) OR 'medication compliance'/exp OR 'medication compliance' OR (('medication'/exp OR medication) AND ('compliance'/exp OR compliance)) OR 'medication non-compliance' OR (('medication'/exp OR medication) AND 'non compliance') OR 'compliance'/exp OR compliance OR 'adherence'/exp OR adherence) AND (((((pharmaceutic* OR pharmaceutical) AND preparation* OR pharmaceutical) AND product* OR drug* OR pharmaceutical) AND regime* OR regime* OR 'oral therapy' OR pill* OR tablet* OR 'capsule'/exp OR capsule OR immunosuppressive) AND agent* OR immunosuppressant* OR nonimmunosuppressant* OR 'non immunosuppressant*') AND (('graft vs host disease'/exp OR 'graft vs host disease' OR (('graft'/exp OR graft) AND vs AND ('host'/exp OR host) AND ('disease'/exp OR disease)) OR 'graft versus host') AND disease* OR 'graft versus host' OR (('graft'/exp OR graft) AND versus AND ('host'/exp OR host) AND disease*) OR 'graft vs host') AND disease*
#5	Medication adherence/MNA and infections	(('stem cell'/exp OR 'stem cell' OR (('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'bone marrow'/exp OR 'bone marrow' OR (('bone'/exp OR bone) AND ('marrow'/exp OR marrow) AND transplant*) OR 'haematopoietic stem cell'/exp OR 'haematopoietic stem cell' OR (haematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'hematopoietic stem cell'/exp OR 'hematopoietic stem cell' OR (hematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR allogeneic) AND transplant* OR haploidentical) AND transplant* AND ('medication adherence'/exp OR 'medication adherence' OR (('medication'/exp OR medication) AND ('adherence'/exp OR adherence)) OR 'adherence medication' OR (('adherence'/exp OR adherence) AND ('medication'/exp OR medication)) OR 'medication nonadherence' OR (('medication'/exp OR medication) AND nonadherence) OR 'medication non-adherence' OR (('medication'/exp OR medication) AND 'non adherence') OR 'medication non adherence' OR (('medication'/exp OR medication) AND non AND ('adherence'/exp OR adherence)) OR 'drug adherence'/exp OR 'drug adherence' OR (('drug'/exp OR drug) AND ('adherence'/exp OR adherence)) OR 'medication persistence'/exp OR 'medication persistence' OR (('medication'/exp OR medication) AND ('persistence'/exp OR persistence)) OR 'medication noncompliance' OR (('medication'/exp OR medication) AND noncompliance) OR 'medication compliance'/exp OR 'medication compliance' OR (('medication'/exp OR medication) AND ('compliance'/exp OR compliance)) OR 'medication non-compliance' OR (('medication'/exp OR medication) AND 'non compliance') OR 'compliance'/exp OR compliance OR 'adherence'/exp OR adherence) AND (((((pharmaceutic* OR pharmaceutical) AND preparation* OR pharmaceutical) AND product* OR drug* OR pharmaceutical) AND regime* OR regime* OR 'oral therapy' OR pill* OR tablet* OR 'capsule'/exp OR capsule OR immunosuppressive) AND agent* OR immunosuppressant* OR nonimmunosuppressant* OR 'non immunosuppressant*') AND ((((infection* OR viral) AND infection* OR bacterial) AND infection* OR fungal) AND infection* OR sepsis*)

#6	Medication adherence/MNA and hospital readmission	('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw) AND ('patient readmission*':ti,ab,kw OR 'unplanned readmission*':ti,ab,kw OR 'hospital readmission*':ti,ab,kw OR rehospitalization:ti,ab,kw OR 'intensive care unit admission*':ti,ab,kw OR 'intensive care unit':ti,ab,kw OR 'emergency room visit*':ti,ab,kw)
#7	Medication adherence/MNA and survival	('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw) AND (survival:ti,ab,kw OR survivor*:ti,ab,kw OR mortality:ti,ab,kw OR death:ti,ab,kw OR 'mortality rate*':ti,ab,kw OR 'progression-free survival':ti,ab,kw)
#8	Medication adherence/MNA and disease relapse	(('stem cell'/exp OR 'stem cell' OR (('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'bone marrow'/exp OR 'bone marrow' OR (('bone'/exp OR bone) AND ('marrow'/exp OR marrow) AND transplant*) OR 'haematopoietic stem cell'/exp OR 'haematopoietic stem cell' OR (haematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'hematopoietic stem cell'/exp OR 'hematopoietic stem cell' OR (hematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR allogeneic) AND transplant* OR haploidentical) AND transplant* AND ('medication adherence'/exp OR 'medication adherence' OR (('medication'/exp OR medication) AND ('adherence'/exp OR adherence)) OR 'adherence medication' OR (('adherence'/exp OR adherence) AND ('medication'/exp OR medication)) OR 'medication nonadherence' OR (('medication'/exp OR medication) AND nonadherence) OR 'medication non-adherence' OR (('medication'/exp OR medication) AND 'non adherence') OR 'medication non adherence' OR (('medication'/exp OR medication) AND non AND ('adherence'/exp OR adherence)) OR 'drug adherence'/exp OR 'drug adherence' OR (('drug'/exp OR drug) AND ('adherence'/exp OR adherence)) OR 'medication persistence'/exp OR 'medication persistence' OR (('medication'/exp OR medication) AND ('persistence'/exp OR persistence)) OR 'medication noncompliance' OR (('medication'/exp OR medication) AND noncompliance) OR 'medication compliance'/exp OR 'medication compliance' OR (('medication'/exp OR medication) AND ('compliance'/exp OR compliance)) OR 'medication non-compliance' OR (('medication'/exp OR medication) AND 'non compliance') OR 'compliance'/exp OR compliance OR 'adherence'/exp OR adherence) AND (((((pharmaceutic* OR pharmaceutical) AND preparation* OR pharmaceutical) AND product* OR drug* OR pharmaceutical) AND regime* OR regime* OR 'oral therapy' OR pill* OR tablet* OR 'capsule'/exp

		OR capsule OR immunosuppressive) AND agent* OR immunosuppressant* OR nonimmunosuppressant* OR 'non immunosuppressant*') AND (recurrence* OR recrudescence* OR relapse* OR 'disease'/exp OR disease) AND relapse*

GvHD, Graft-versus-Host Disease; MNA, medication non-adherence.

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Medication adherence among allogeneic haematopoietic stem cell transplant recipients: a systematic review protocol

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ABSTRACT

Introduction Patients receiving a haematopoietic stem cell transplant (HSCT) are subjected to complex oral medications based on prophylactic and immunosuppressive treatments. Adherence to medication plays a role in survival, and medication non-adherence (MNA) is closely associated with graft-versus-host disease (GvHD) and other complications. The aim of this systematic review is to summarise the available evidence regarding prevalence rates of medication adherence, the risk factors of MNA, the effectiveness of interventions to increase medication adherence and the outcomes associated with MNA.

Methods and analysis We designed a systematic review according to the Joanna Briggs Institute methodology. We will search the Cochrane Library and the CINAHL, EMBASE, MEDLINE via PubMed, PsychINFO and Scopus databases. We will include published and unpublished primary studies: (a) on humans, from inception until 10 May 2022; (b) written in any language; (c) experimental (randomised and non-randomised), observational (prospective, retrospective cohort and case-control), correlational, cross-sectional and longitudinal; and (d) with a low risk of bias, according to the quality assessment we perform. We will exclude secondary and qualitative studies, protocols, publications without original data, including paediatrics or related to autologous HSCT. The primary outcome will be the prevalence of oral medication adherence; the secondary outcomes will be the risk factors of MNA, the interventions aimed at increasing medication adherence and the outcomes of MNA. Two researchers will independently screen the eligible studies, then extract and describe the data. Disagreements will be resolved by a third researcher. We will provide a qualitative narrative synthesis of the findings.

Ethics and dissemination Ethical approval is not required given that previously published studies will be utilised. We will disseminate the findings through conference presentations and publications in international peer-reviewed scientific journals.

PROSPERO registration number CRD42022315298.

Keywords: Graft-versus-host disease, Haematopoietic stem cell transplantation, Immunosuppressants, Infections, Interventions, Medication adherence, Medication non-adherence, Oral therapy, Prophylaxis, Risk factors

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ARTICLE SUMMARY

Strengths and limitations of this study

- The review will examine six databases and grey literature, leading to a broad synthesis of the available evidence regarding medication adherence among adult recipients of an allogeneic haematopoietic stem cell transplant.
- The review will include primary studies without language and time restrictions, allowing the maximum inclusion of studies published in the field.
- Studies with low methodological quality will be excluded, thus ensuring rigor and robustness in the findings of the review.
- The independence of the researchers in the selection, analysis and summary of the studies will ensure rigor and strengthen the findings of the review.
- According to the expected heterogeneity across studies, a quantitative synthesis of the results in the form of meta-analysis may not be possible.

INTRODUCTION

The number of haematopoietic stem cell transplants (HSCTs) is increasing worldwide, with 48,512 transplants in the year 2019, including 19,798 allogeneic transplants (41%) in Europe,[1] and 23,768 HSCTs in the United States, of which 9,498 (40%) were allogeneic.[2] Patients who receive a HSCT must take a complex oral medication regimen based upon prophylactic and immunosuppressive treatment.[3-5] Thus, patients may encounter difficulties managing the high number of tablets, which may vary over time due to medication prescription changes.[6] However, a high level of adherence to this very complex regimen is required to avoid infections and to minimise the risk of other complications, such as graft-versus-host disease (GvHD) or disease relapse.[7] It is well known that adherence plays a key role during and after the HSCT, increasing survival, and medication non-adherence (MNA) is closely associated with GvHD.[4,6,8,9]

Adherence is defined by the World Health Organization (WHO) as 'the extent to which the patient follows medical instructions'.[10] It is composed of three phases: initiation (starting the prescribed medication intake), implementation (the agreement between a patient's actual dosing compared with the medical prescription) and persistence (the length of time between the first and last dose taken).[11] Differently, the term

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'compliance' implies patient passivity;[12,13] therefore, the adherence concept and term was considered as a point of reference in this study.

The adherence rate in developed countries is about 50% among patients with chronic illness.[13] To the best of our knowledge, the only systematic review available regarding medication adherence among recipients of a HSCT was published by Morrison and colleagues in 2017.[7] Authors included five studies, two in adults,[14,15] one in both children and adults[16] and two in paediatric population.[6,17] The estimated adherence to oral therapy in that review was between 33% and 94.7%[7] and it has been reported to decline over time among adults,[15,16] except for Chieng *et al.*,[14] who reported an improvement in medication adherence over time. More recently, a prospective survey conducted by Ice and colleagues[4] showed that out of 200 patients, 51% of them were not adherent to non-immunosuppressant medications, and 37.9% were not adherent to oral immunosuppressants. In a cross-sectional multicentric study enrolling 203 French adult recipients of an allogeneic HSCT, the MNA rate was 75%.[18] In addition, in a secondary analysis of long-term survivors after an allogeneic HSCT, the authors reported a MNA prevalence rate of 68.7% to immunosuppressants.[8]

Various methods have been used to assess medication adherence, which can be categorised into subjective, objective or biochemical measures.[12] Among the first category, self-reported questionnaires (i.e. the Compliance Evaluation Test,[18] the modified Self-Efficacy for Appropriate Medication Use Scale,[19] the Immunosuppressant Therapy Adherence Scale[4] and the Morisky Medication Adherence Scale[4,7]) have been used. Among the objective measures, the electronic Medication Administration Record[19] and the prescription refill records[4] have been adopted by researchers, while among biochemical measures, the serum drug level measurements of immunosuppressants, alone[20] or in combination, have been used.[4] According to the available evidence, the use of self-report approaches combined with objective methods has been recommended[7,12,13] to prevent heterogeneity in prevalence rates across studies.[21] Moreover, it has been recommended to report when adherence is measured, the medications prescribed and the conceptual definition of adherence considered by researchers to increase the methodological quality of studies.[21]

The WHO[13] has classified reasons for poor medication adherence in chronic illnesses into five categories: socioeconomic factors, health system/health care team–related factors, disease-related factors, therapy-related factors and patient-related factors. Among individuals who have undergone solid transplantation, several factors have been investigated regarding their association with MNA.[3] At the patient-related level, some studies have documented the role of psychosocial factors[9,22] by using the Stanford Integrated Psychosocial Assessment for Transplant recipients (SIPAT), previously validated in the population who had received a solid organ transplant,[22,23] and the Transplant Evaluation Rating Scale (TERS) as a

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pretransplant screening instrument to predict mortality, medication adherence and GvHD occurrence. Patient age,[4] education,[24] distress[4] as well as the efficacy of the caregiver(s)[25] have also been investigated. At the therapy-related level, the long-term duration of the therapy[25] as well as the higher number of daily immunosuppressant pills have been studied.[8] However, as has emerged from a published systematic review[7] and recommended by the Patient-Centered Outcomes Working Group of the National Institutes of Health in 2015,[26] future research is warranted to understand the facilitators and barriers of medication adherence.

Interventions to improve medication adherence have only been investigated to a limited extent, and to our best knowledge no summary of the available studies has been published. Specifically, a recent French retro-prospective study[20] has assessed the effectiveness of pharmaceutical consultations on the adherence to oral immunosuppressive therapy. The intervention was performed by a hospital pharmacist to patients who had received an allogeneic HSCT (n = 26) the day before discharge from the bone marrow transplantation centre (BMTC) and from 2 to 4 weeks after discharge. There were no significant findings in favour of the intervention group on medication adherence. In their quasi-experimental study, Polito *et al.*[19] compared medication counselling by a pharmacist and self-management with nursing supervision until discharge to a control group among 51 recipients of an allogeneic HSCT. There were no differences between the two groups regarding medication adherence, although at discharge the experimental group had higher median knowledge scores. Moreover, evidence regarding different educational strategies, such as peer education or the involvement of patients as trainers, has not yet been summarised.

The effects of medication adherence on clinical outcomes, such as infections, GvHD, disease relapse and mortality, are emerging in the literature.[4,8,9,20,22] Mishkin *et al.*[22] reported no significant associations between SIPAT ratings and survival; however, patients with a high-risk SIPAT score had an increased risk of being admitted to the intensive care unit (ICU). Similarly, no differences emerged in survival between adherent and non-adherent patients in the prospective survey by Ice and colleagues,[4] nor with respect to acute GvHD and infections in the quasi-experimental trial by Charra *et al.*[20] On the other hand, there was a difference in the development of mild chronic GvHD among non-adherent patients compared with adherent patients.[4] While pre-transplant screening with the TERS has been documented to contribute to prediction of survival after HSCT, the TERS score did not correlate with the MESI – used as an adherence measure – suggesting that mortality is not associated with MNA.[9] There is a need to summarise the evidence regarding clinical outcomes to discriminate between adherence and non-adherence behaviour and to identify patients at increased risk[21] and in need of tailored interventions.

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Studies regarding oral medication adherence prevalence, its associated factors, the effectiveness of interventions promoting it and the associated clinical outcomes among the adult recipients of an allogeneic HSCT are sparse and date back to recent years.[4,7-9,18-20,22] Therefore, this review is intended to provide a comprehensive and quality-assessed summary of studies to inform clinical practice and to address further primary research. The findings will also increase the awareness of patients and their caregivers regarding MNA and its consequences, allowing them to act at any modifiable risk factors with targeted interventions. Our primary aim is to summarise the available evidence regarding the phenomenon of oral medication adherence among adult recipients of an allogeneic HSCT based on prevalence rates and data collection tools used to measure it. Our secondary aim is to summarise the evidence regarding the predictors and risk factors of MNA, the effectiveness of interventions to promote medication adherence, and the clinical outcomes influenced by MNA, as documented in primary studies to date.

METHODS AND ANALYSIS

We have registered the present protocol in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number CRD42022315298, on 8 April 2022). This protocol follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P) (see Supplementary Table 1).[27] The review process will be completed within 6 months of protocol registration, and the systematic review will be reported following the updated Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement.[28]

Eligibility criteria

We will consider all primary studies to be eligible for the review, ensuring the following elements as defined with the Population, Intervention, Comparison and Outcomes and Time – PICO(T) – framework, as recommended by the Joanna Briggs Institute (JBI) methodology[29]:

- *Population* – adult (age ≥ 18 years old) patients who have undergone allogeneic or haploidentical HSCT and must take oral medication in the post-transplant phase, after discharge from BMTCs, regardless of the number of pills prescribed.
- *Intervention* – all kinds of interventions aimed at increasing medication adherence delivered from any health care providers (HCPs), during hospitalisation in the BMTCs and/or in the post-transplant phase, as an outpatient or remote follow-up.

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- *Comparison* – usual care (i.e. consultations, counselling, peer-education and/or no comparison group).
- *Outcome(s)* – primary outcome describing the prevalence of oral medication adherence to immunosuppressant and non-immunosuppressant medications among the selected population after the discharge from BMTCs. The secondary outcomes are (a) risk factors, barriers and predictors, and possibly related MNA assessment tools; (b) effectiveness of interventions aimed at increasing medication adherence; and (c) infections (bacterial, viral and fungal), acute and chronic GvHD, hospital readmission rate, disease relapse and overall survival, as associated with MNA.
- *Time* – from transplantation to 4 years post-HSCT, according to the longest median follow-up among the available studies.[8]

Inclusion and exclusion criteria

Our target population consists of adult patients who have undergone allogeneic or haploidentical HSCT, for any transplantation indications, and must take oral medications (both immunosuppressants and non-immunosuppressants, such as prophylactic medications) in the post-transplant phase.

We will include all published and unpublished primary studies: (a) on humans, from inception until 10 May 2022; (b) written in any language; (c) experimental (randomised and non-randomised), observational (prospective, retrospective cohort and case-control), correlational, cross-sectional and longitudinal, in addition to case series and case reports; and d) with a low risk of bias, according to the quality assessment findings.[29] Therefore, we will exclude secondary studies, qualitative studies, study protocols, case series and case reports and publications without original data (i.e. comments, letters to the editor and editorials) or including the paediatric population (< 18 years old) or relating to autologous HSCT. In addition, we will exclude studies where adherence does not relate to medication (i.e. diet regimen, physical exercise).

Primary and secondary outcomes

The primary outcome of the review is the oral medication adherence rate. We will adopt the theoretical framework proposed by the WHO,[13] where adherence is defined as 'the extent to which a person's behaviour – taking medication, following a diet, and/or executing lifestyle changes, corresponds with agreed recommendations from a health care provider'. However, the focus of adherence will be the medication

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regimen, thus excluding 'a diet, and/or executing lifestyle changes' because they are not in line with our research aims. To measure the medication adherence rate, we will investigate subjective (i.e. self-report questionnaires), objective (e.g. prescription refill records) or biochemical measurements (serum drug levels).[12]

The secondary outcomes of this review are the following:

(a) risk factors, barriers or predictors of MNA as measured with tools.

(b) interventions increasing medication adherence (i.e. full description, frequency of provision and duration, comparator).

(c) clinical outcomes as associated with MNA, such as the incidence rates of bacterial, viral and fungal infections; the incidence rates of acute and chronic GvHD[30]; hospital and ICU readmission prevalence rates; disease relapse rates; and overall survival rates. We will assess correlations or associations, if any, between medication adherence/MNA and patient-reported outcomes (i.e. health-related quality of life, symptom experiences and patient satisfaction).[31]

Information source

We will access the following databases: the Cochrane Library, Cumulative Index of Nursing and Allied Health (CINAHL), EMBASE, MEDLINE via PubMed, PsychINFO and Scopus. An initial search in the PubMed database will be performed by the first researcher (CV) to identify keywords and Medical Subject Headings (MESH) appropriate to the aims of the review. Then, the search strategy will be refined and tested by the other researchers (IM, AP). Thereafter, the search string will be inserted into the other databases.

We will also search for grey literature, including dissertation theses and reports, protocols or guidelines from international scientific societies regarding medication adherence in the HSCT population. We will screen the reference lists of the included articles and the excluded systematic reviews using a hand search, to identify additional sources. If the full text of a study is not available, we will contact the authors directly. In the case of studies written in languages other than English, Italian and German, all accessible by authors, we will employ native translators to provide an English translation.

Search strategy

Different search strings have been designed and applied, considering the primary outcome and the three secondary outcomes. The date of the last update of the literature search was 10 May 2022. The final search

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strategies for the six selected databases, with MeSH terms and keywords, are shown in Supplementary Table 2. No filters and/or limits have been used.

Data management and selection process

We will download citations retrieved from the six electronic databases and import them into software (EndNote®) that will facilitate the study selection process, with automatic identification and exclusion of the duplicates.

After the removal of duplicates, the first researcher (CV) will screen article titles and abstracts according to the inclusion and exclusion criteria; then, another researcher (IM) will independently screen the articles, reaching a consensus on eligible studies to include in the review. Disagreements between the researchers will be solved using consensus and consultation with a third researcher (AP), who will not have participated in the screening phase. A full-text assessment will then be performed by two researchers (CV, IM) independently, and then with common agreement with the third researcher (AP). The lists of references from the studies will also be screened, with the aim of retrieving and adding relevant studies, following the inclusion criteria. We will present the selection process in the PRISMA flow-chart.[28]

Data collection process

A pilot-grid will be created in Microsoft Word® to extract data for at least 10% of the included articles, conducted independently by the first (CV) and second (IM) authors. After the pilot test, changes will be included if necessary, and the final chart will be developed, where study data will be reported and double-checked. Discrepancies in data extraction by the researchers will be discussed until a consensus is reached, with the supervision of a third researcher (AP).

An example of information that will be extracted from each study could be:

- main characteristics (author[s], publication year, country, study design, study setting, study duration);
- main and secondary study aims(s);
- population profile, such as the number of participants, median age, gender distribution, race, ethnicity, socioeconomic status (where available), haematological disease due to HSCT, type of allogeneic HSCT, time from HSCT and name of oral medication(s), categorised into immunosuppressants or non-immunosuppressants.

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- main results of the study according to the primary and the secondary outcome of the review:
  (a) risk factors, predictors, barriers (description), by also describing the tools used to assess the risk of MNA;
  (b) intervention(s) to improve medication adherence (description of the intervention, timing and provider[s] of the intervention[s], comparator);
  (c) outcomes (i.e. incidence rate of bacterial, viral and/or fungal infections, acute and/or chronic GvHD, hospital and/or ICU readmission rate, disease relapse rate and/or overall survival rate) and time from HSCT (in days) to assess them.

Risk of bias

We will perform a risk of bias assessment of all the individual studies included. The assessment will be performed independently by two researchers (CV, IM), and conflicts will be solved by a third researcher (AP).

The assessment will be based on the use of the JBI Appraisal Checklists,[29] according to the study design of the included studies. The JBI has different checklists for each study design (i.e. randomised controlled trials, quasi-experimental, cohort). An example is the Critical Appraisal Instrument for Studies Reporting Prevalence Data that will be used to assess the risk of bias of cross-sectional studies, which assesses the appropriateness of inclusion criteria (target population, sampling and sample size), the description of participants and the setting, the use of valid and reliable measurement of the studied condition, the use of appropriate statistical analysis, and the assessment and management of adequate response rate.[32]

There are four possible responses for each item: 'yes', 'no', 'unclear' or 'not applicable'. The overall appraisal for each included study will consist of 'seek further info' or 'include' or 'exclude' the study from the final inclusion.[29] Therefore, studies assessed as excludable will not be considered in the data synthesis and a brief comment explaining the reason for the exclusion will be indicated. We will provide a table reporting the results of the risk of bias assessment for each included study.

Data synthesis

Two researchers (CV, IM) will independently summarise the findings, including only those studies considered eligible after the risk of bias assessment. First, the main study features will be summarised by reporting, for example, the origin country of the authors, the study aims and design, its duration, the sample size, and the participants (i.e. age, gender, ethnicity, education, type of HSCT). Then, a qualitative narrative synthesis of

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the results will be provided; specifically, the findings that emerge will be synthesised in four dimensions, according to the primary and the three secondary aims of this review. For example, the prevalence rates reported in the studies for immunosuppressant and non-immunosuppressant medications will be presented with the minimum, maximum and median values. The data collection methods, the description of risk factors of MNA and the interventions applied to improve adherence will also be summarised.

Regarding the investigated prevalence and outcomes, due to the expected different medication adherence measures and interventions across studies,[7] the heterogeneity of the identified studies may prevent the ability to conduct a meta-analysis. However, if the data allow, two researchers (CV, IM) will independently investigate the heterogeneity of the studies using the I^2 statistic.

Confidence in cumulative evidence

We will assess the quality of the evidence by using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework[33] for both primary and secondary outcomes. 'High', 'moderate', 'low' and 'very low' are the levels that will be used to assess the strength and quality of the evidence regarding the identified outcomes. The evaluation will be performed independently by two researchers (CV, IM), and an expert researcher (AP) will resolve any disagreements.

Patient and public involvement

None.

ETHICS AND DISSEMINATION

Because the study will only be based on published and retrievable literature, no ethics approval is required. We will submit the findings of our systematic review to international peer-reviewed scientific journals based on the identified primary and secondary outcomes. Furthermore, we will disseminate a summary of the systematic review through professional meetings and conference presentations, to promote implementation in HCP practice, such as providing screening tools to recognise patients at risk, barriers and risk factors of MNA or interventions to improve medication adherence.

DISCUSSION

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We aim to summarise the available studies regarding the phenomenon of oral medication adherence by means of prevalence rates, predictors and risk factors of MNA, the effectiveness of interventions to support it and outcomes associated with MNA among the HSCT population. Although we will employ a systematic approach and search an extensive number of databases, a publication bias might occur. Moreover, the expected heterogeneity across studies regarding their measures, tools and interventions[7] might prevent a quantitative synthesis of the results. Furthermore, the external validity of the findings will be only for adult population: we decided to exclude the paediatric population because of the differences in psychological answers to the disease and medication posology[21] and to the influence of the dyad child-parent on medication adherence.[7,25] In addition, facilitators and barriers could be different between the paediatric and the adult population.[7]

The findings will be of interest to various stakeholders, including HCPs (such as nurses, haematologists and psychologists working in BMTCs and in the follow-up setting), patients who have received a HSCT, their caregivers and their representatives. Although disease relapse remains the leading cause of mortality among patients after receiving a HSCT, a substantial improvement in survival and a reduction of complications have been reported over the years.[34] Therefore, also considering the increase in the incidence of GvHD,[8] medication adherence has become a priority in designing and delivering health care interventions during and after HSCT.[4,6,8,9]

Complications or failure of transplant during hospitalisation have mostly been attributed to biological reasons.[9] There has been a recent increase in attention towards medication adherence among patients who have received a transplant, especially in the case of early discharge and after, in the outpatient follow-up. Therefore, the findings will increase the information available to HCPs, patients, their caregivers and representatives, in understanding the factors hindering or promoting medication adherence. Moreover, this review will inform HCPs regarding the most effective interventions capable of promoting medication adherence by considering all factors involved – from those related to the complexity of the therapy to the uncertainty and psychological distress experienced by the patients and their caregivers.[35] Providing data on the phenomenon of medication adherence might increase the awareness of HCPs regarding the educational needs during hospitalisation and over time, because not all BMTCs have reference HPCs in charge of transitional care. Moreover, providing evidence might increase the opportunities to benchmark the data at each centre with that accumulated in the literature and appropriately summarised. Finally, the findings of this review may also reveal possible gaps in the HSCT literature and highlight research questions to address with primary studies.

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Contributors

CV and AP conceptualized the protocol. CV wrote the protocol. IM and AP methodologically and critically revised the protocol. CV is the guarantor of the review. All authors have contributed, read and approved the final protocol.

Competing interests

None declared.

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Patient consent for publication

Not required.

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Supplementary table 1 Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P) checklist: recommended items to address in a systematic review protocol.[27]

Section/topic	#	Checklist item	Information reported		Section where item is reported
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	Yes		Title
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		No	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number in the Abstract	Yes		Abstract and Methods and Analysis
Authors					
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Yes		First page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Yes		Author contribution
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		No	
Support					
Sources	5a	Indicate sources of financial or other support for the review		No	
Sponsor	5b	Provide name for the review funder and/or sponsor		No	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		No	
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	Yes		Introduction
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Yes		Eligibility criteria
METHODS					
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Yes		Inclusion and exclusion criteria
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Yes		Information sources
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Yes		Search strategy
STUDY RECORDS					

Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Yes		Data management and selection process
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Yes		Data management and selection process
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Yes		Data collection process
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications		No	
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Yes		Primary and secondary outcomes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Yes		Risk of bias
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	Yes		Data synthesis
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Yes		Data synthesis
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)		No	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Yes		Data synthesis
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)		No	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Yes		Confidence in cumulative evidence

n, number.

Supplementary table 2 Final search strategy for the six investigated databases (no limits and/or filters have been used).

Database	String number	Researched outcome	Search strings
PUBMED	#1	Medication adherence/MNA	("Hematopoietic Stem Cell Transplantation"[Mesh] OR "Transplantation, Homologous"[Mesh] OR stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND ("Medication Adherence"[Mesh] OR medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR "Pharmaceutical Preparations"[Mesh] OR pharmaceutical regime* OR regime* OR "oral therapy" OR pill* OR tablet* OR capsule OR "Immunosuppressive Agents"[Mesh] OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND ("Risk Factors"[Mesh] OR risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR "screening tool" OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND ("Psychosocial Intervention"[Mesh] OR intervention* OR psychological intervention* OR "Patient Education as Topic"[Mesh] OR "patient education" OR education OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND ("Graft vs Host Disease"[Mesh] OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND ("Infections"[Mesh] OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND ("Patient Readmission"[Mesh] OR patient readmission* OR unplanned readmission OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND ("Survival"[Mesh] OR "Mortality"[Mesh] OR survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)
	#8	Medication adherence/MNA and disease relapse	#1 AND ("Recurrence"[Mesh] OR recurrence* OR recrudescence* OR relapse* OR disease relapse*)
CINAHL	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant* OR (MH "Bone Marrow Transplantation") OR (MH "Bone Marrow Transplantation, Allogeneic")) AND ((MH "Medication Compliance") OR (MH "Compliance with Medical Regimen (Saba CCC)") OR (MH "Compliance with Medication Regimen (Saba CCC)" OR medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR "oral therapy" OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR nonimmunosuppressant* OR non-immunosuppressant* OR (MH "Immunosuppressive Agents") OR (MH "Immunosuppression"))

	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR "screening tool" OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier* OR (MH "Risk Factors") OR (MH "Social Factors"))
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication OR (MH "Psychosocial Intervention") OR (MH "Nursing Interventions") OR (MH "Intervention Trials") OR (MH "Patient Education") OR (MH "Counseling"))
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease* OR (MH "Graft Versus Host Disease"))
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis* OR (MH "Sepsis") OR (MH "Infection"))
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit* OR (MH "Readmission") OR (MH "Intensive Care Units"))
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival OR (MH "Survival") OR (MH "Mortality") OR (MH "Death"))
	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse* OR (MH "Recurrence"))
Cochrane Library	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND (medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR "oral therapy" OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR non immunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR "screening tool" OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)
	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse*)
PsychINFO	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND (medication

			adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR oral therapy OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR non immunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR screening tool OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)
	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse*)
SCOPUS	#1	Medication adherence/MNA	(stem cell transplant* OR bone marrow transplant* OR haematopoietic stem cell transplant* OR hematopoietic stem cell transplant* OR allogeneic transplant* OR haploidentical transplant*) AND (medication adherence OR adherence medication OR medication nonadherence OR medication non-adherence OR medication non adherence OR Drug Adherence OR Medication Persistence OR medication noncompliance OR medication compliance OR medication non-compliance OR compliance OR adherence) AND (pharmaceutic* OR pharmaceutical preparation* OR pharmaceutical product* OR drug* OR pharmaceutical regime* OR regime* OR “oral therapy” OR pill* OR tablet* OR capsule OR Immunosuppressive Agent* OR immunosuppressant* OR non immunosuppressant* OR non-immunosuppressant*)
	#2	MNA and risk factor	(#1 OR pre-transplant*) AND (risk factor* OR social risk factor* OR health correlate* OR risk score* OR risk factor score* OR predictor* OR screening OR “screening tool” OR psychosocial assessment* OR evaluation* OR psychosocial* OR facilitate* OR barrier*)
	#3	Interventions increasing medication adherence	#1 AND (Psychosocial Intervention* OR intervention* OR psychological intervention* OR patient education OR education* OR consultation* OR counselling* OR implementation OR self-medication)
	#4	Medication adherence/MNA and GvHD	#1 AND (Graft vs Host Disease OR graft-versus-host disease* OR graft-versus-host OR graft versus host disease* OR graft-vs-host disease*)
	#5	Medication adherence/MNA and infections	#1 AND (Infection* OR viral infection* OR bacterial infection* OR fungal infection* OR sepsis*)
	#6	Medication adherence/MNA and hospital readmission	#1 AND (patient readmission* OR unplanned readmission* OR hospital readmission* OR rehospitalization OR intensive care unit admission* OR intensive care unit OR emergency room visit*)
	#7	Medication adherence/MNA and survival	#1 AND (survival OR survivor* OR mortality OR death OR mortality rate* OR progression-free survival)

	#8	Medication adherence/MNA and disease relapse	#1 AND (recurrence* OR recrudescence* OR relapse* OR disease relapse*)
EMBASE	#1	Medication adherence/MNA	('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw)
	#2	MNA and risk factor	('risk factor*':ti,ab,kw OR 'social risk factor*':ti,ab,kw OR 'health correlate*':ti,ab,kw OR 'risk score*':ti,ab,kw OR 'risk factor score*':ti,ab,kw OR predictor*:ti,ab,kw OR screening:ti,ab,kw OR 'screening tool':ti,ab,kw OR 'psychosocial assessment*':ti,ab,kw OR evaluation*:ti,ab,kw OR psychosocial*:ti,ab,kw OR facilitate*:ti,ab,kw OR barrier*:ti,ab,kw) AND ('pre transplant*':ti,ab,kw OR 'stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw)
	#3	Interventions increasing medication adherence	('psychosocial intervention*':ti,ab,kw OR intervention*:ti,ab,kw OR 'psychological intervention*':ti,ab,kw OR 'patient education':ti,ab,kw OR education*:ti,ab,kw OR consultation*:ti,ab,kw OR counselling*:ti,ab,kw OR implementation:ti,ab,kw OR 'self medication':ti,ab,kw) AND ('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw)
	#4	Medication adherence/MNA and GvHD	(('stem cell'/exp OR 'stem cell' OR (('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'bone marrow'/exp OR 'bone marrow' OR (('bone'/exp OR bone) AND ('marrow'/exp OR marrow) AND transplant*) OR 'haematopoietic stem cell'/exp OR 'haematopoietic stem cell' OR (haematopoietic AND

		('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'hematopoietic stem cell'/exp OR 'hematopoietic stem cell' OR (hematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR allogeneic) AND transplant* OR haploidentical) AND transplant* AND ('medication adherence'/exp OR 'medication adherence' OR (('medication'/exp OR medication) AND ('adherence'/exp OR adherence)) OR 'adherence medication' OR (('adherence'/exp OR adherence) AND ('medication'/exp OR medication)) OR 'medication nonadherence' OR (('medication'/exp OR medication) AND nonadherence) OR 'medication non-adherence' OR (('medication'/exp OR medication) AND 'non adherence') OR 'medication non adherence' OR (('medication'/exp OR medication) AND non AND ('adherence'/exp OR adherence)) OR 'drug adherence'/exp OR 'drug adherence' OR (('drug'/exp OR drug) AND ('adherence'/exp OR adherence)) OR 'medication persistence'/exp OR 'medication persistence' OR (('medication'/exp OR medication) AND ('persistence'/exp OR persistence)) OR 'medication noncompliance' OR (('medication'/exp OR medication) AND noncompliance) OR 'medication compliance'/exp OR 'medication compliance' OR (('medication'/exp OR medication) AND ('compliance'/exp OR compliance)) OR 'medication non-compliance' OR (('medication'/exp OR medication) AND 'non compliance') OR 'compliance'/exp OR compliance OR 'adherence'/exp OR adherence) AND (((((pharmaceutic* OR pharmaceutical) AND preparation* OR pharmaceutical) AND product* OR drug* OR pharmaceutical) AND regime* OR regime* OR 'oral therapy' OR pill* OR tablet* OR 'capsule'/exp OR capsule OR immunosuppressive) AND agent* OR immunosuppressant* OR nonimmunosuppressant* OR 'non immunosuppressant*') AND (('graft vs host disease'/exp OR 'graft vs host disease' OR (('graft'/exp OR graft) AND vs AND ('host'/exp OR host) AND ('disease'/exp OR disease)) OR 'graft versus host') AND disease* OR 'graft versus host' OR (('graft'/exp OR graft) AND versus AND ('host'/exp OR host) AND disease*) OR 'graft vs host') AND disease*
#5	Medication adherence/MNA and infections	(('stem cell'/exp OR 'stem cell' OR (('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'bone marrow'/exp OR 'bone marrow' OR (('bone'/exp OR bone) AND ('marrow'/exp OR marrow) AND transplant*) OR 'haematopoietic stem cell'/exp OR 'haematopoietic stem cell' OR (haematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'hematopoietic stem cell'/exp OR 'hematopoietic stem cell' OR (hematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR allogeneic) AND transplant* OR haploidentical) AND transplant* AND ('medication adherence'/exp OR 'medication adherence' OR (('medication'/exp OR medication) AND ('adherence'/exp OR adherence)) OR 'adherence medication' OR (('adherence'/exp OR adherence) AND ('medication'/exp OR medication)) OR 'medication nonadherence' OR (('medication'/exp OR medication) AND nonadherence) OR 'medication non-adherence' OR (('medication'/exp OR medication) AND 'non adherence') OR 'medication non adherence' OR (('medication'/exp OR medication) AND non AND ('adherence'/exp OR adherence)) OR 'drug adherence'/exp OR 'drug adherence' OR (('drug'/exp OR drug) AND ('adherence'/exp OR adherence)) OR 'medication persistence'/exp OR 'medication persistence' OR (('medication'/exp OR medication) AND ('persistence'/exp OR persistence)) OR 'medication noncompliance' OR (('medication'/exp OR medication) AND noncompliance) OR 'medication compliance'/exp OR 'medication compliance' OR (('medication'/exp OR medication) AND ('compliance'/exp OR compliance)) OR 'medication non-compliance' OR (('medication'/exp OR medication) AND 'non compliance') OR 'compliance'/exp OR compliance OR 'adherence'/exp OR adherence) AND (((((pharmaceutic* OR pharmaceutical) AND preparation* OR pharmaceutical) AND product* OR drug* OR pharmaceutical) AND regime* OR regime* OR 'oral therapy' OR pill* OR tablet* OR 'capsule'/exp OR capsule OR immunosuppressive) AND agent* OR immunosuppressant* OR nonimmunosuppressant* OR 'non immunosuppressant*') AND ((((infection* OR viral) AND infection* OR bacterial) AND infection* OR fungal) AND infection* OR sepsis*)

#6	Medication adherence/MNA and hospital readmission	('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw) AND ('patient readmission*':ti,ab,kw OR 'unplanned readmission*':ti,ab,kw OR 'hospital readmission*':ti,ab,kw OR rehospitalization:ti,ab,kw OR 'intensive care unit admission*':ti,ab,kw OR 'intensive care unit':ti,ab,kw OR 'emergency room visit*':ti,ab,kw)
#7	Medication adherence/MNA and survival	('stem cell transplant*':ti,ab,kw OR 'bone marrow transplant*':ti,ab,kw OR 'haematopoietic stem cell transplant*':ti,ab,kw OR 'hematopoietic stem cell transplant*':ti,ab,kw OR 'allogeneic transplant*':ti,ab,kw OR 'haploidentical transplant*':ti,ab,kw) AND ('medication adherence':ti,ab,kw OR 'adherence medication':ti,ab,kw OR 'medication nonadherence':ti,ab,kw OR 'medication non-adherence':ti,ab,kw OR 'medication non adherence':ti,ab,kw OR 'drug adherence':ti,ab,kw OR 'medication persistence':ti,ab,kw OR 'medication noncompliance':ti,ab,kw OR 'medication compliance':ti,ab,kw OR 'medication non-compliance':ti,ab,kw OR compliance:ti,ab,kw OR adherence:ti,ab,kw) AND (pharmaceutic*:ti,ab,kw OR 'pharmaceutical preparation*':ti,ab,kw OR 'pharmaceutical product*':ti,ab,kw OR drug*:ti,ab,kw OR 'pharmaceutical regime*':ti,ab,kw OR regime*:ti,ab,kw OR 'oral therapy':ti,ab,kw OR pill*:ti,ab,kw OR tablet*:ti,ab,kw OR capsule:ti,ab,kw OR 'immunosuppressive agent*':ti,ab,kw OR immunosuppressant*:ti,ab,kw OR nonimmunosuppressant*:ti,ab,kw OR 'non immunosuppressant*':ti,ab,kw) AND (survival:ti,ab,kw OR survivor*:ti,ab,kw OR mortality:ti,ab,kw OR death:ti,ab,kw OR 'mortality rate*':ti,ab,kw OR 'progression-free survival':ti,ab,kw)
#8	Medication adherence/MNA and disease relapse	(('stem cell'/exp OR 'stem cell' OR (('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'bone marrow'/exp OR 'bone marrow' OR (('bone'/exp OR bone) AND ('marrow'/exp OR marrow) AND transplant*) OR 'haematopoietic stem cell'/exp OR 'haematopoietic stem cell' OR (haematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR 'hematopoietic stem cell'/exp OR 'hematopoietic stem cell' OR (hematopoietic AND ('stem'/exp OR stem) AND ('cell'/exp OR cell) AND transplant*) OR allogeneic) AND transplant* OR haploidentical) AND transplant* AND ('medication adherence'/exp OR 'medication adherence' OR (('medication'/exp OR medication) AND ('adherence'/exp OR adherence)) OR 'adherence medication' OR (('adherence'/exp OR adherence) AND ('medication'/exp OR medication)) OR 'medication nonadherence' OR (('medication'/exp OR medication) AND nonadherence) OR 'medication non-adherence' OR (('medication'/exp OR medication) AND 'non adherence') OR 'medication non adherence' OR (('medication'/exp OR medication) AND non AND ('adherence'/exp OR adherence)) OR 'drug adherence'/exp OR 'drug adherence' OR (('drug'/exp OR drug) AND ('adherence'/exp OR adherence)) OR 'medication persistence'/exp OR 'medication persistence' OR (('medication'/exp OR medication) AND ('persistence'/exp OR persistence)) OR 'medication noncompliance' OR (('medication'/exp OR medication) AND noncompliance) OR 'medication compliance'/exp OR 'medication compliance' OR (('medication'/exp OR medication) AND ('compliance'/exp OR compliance)) OR 'medication non-compliance' OR (('medication'/exp OR medication) AND 'non compliance') OR 'compliance'/exp OR compliance OR 'adherence'/exp OR adherence) AND (((((pharmaceutic* OR pharmaceutical) AND preparation* OR pharmaceutical) AND product* OR drug* OR pharmaceutical) AND regime* OR regime* OR 'oral therapy' OR pill* OR tablet* OR 'capsule'/exp

		OR capsule OR immunosuppressive) AND agent* OR immunosuppressant* OR nonimmunosuppressant* OR 'non immunosuppressant*') AND (recurrence* OR recrudescence* OR relapse* OR 'disease'/exp OR disease) AND relapse*

GvHD, Graft-versus-Host Disease; MNA, medication non-adherence.

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